Chapter 8: Problems with Solutions

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2. A ball of mass m falls from a height h to the floor. (a) Write the appropriate version of Equation 8.2 for the system of the ball and the Earth and use it to calculate the speed of the ball just before it strikes the Earth. (b) Write the appropriate version of Equation 8.2 for the system of the ball and use it to calculate the speed of the ball just before it strikes the Earth.

(a) The system of the ball and the Earth is isolated. The gravitational energy of the system decreases as the kinetic energy increases.

$$\boxed{\Delta K + \Delta U = 0}$$

$$\left(\frac{1}{2}mv^2 - 0\right) + \left(-mgh - 0\right) = 0 \rightarrow \frac{1}{2}mv^2 = mgy$$

$$\boxed{v = \sqrt{2gh}}$$

(b) The gravity force does positive work on the ball as the ball moves downward. The Earth is assumed to remain stationary, so no work is done on it.

$$\Delta K = W$$

$$\left(\frac{1}{2}mv^2 - 0\right) = mgh \rightarrow \frac{1}{2}mv^2 = mgy$$

$$\boxed{v = \sqrt{2gh}}$$

5. AMT M **Review.** A bead slides without friction around a loop-the-loop (Fig. P8.5). The bead is released from rest at a height $h = 3.50R$. (a) What is its speed at point Ⓐ? (b) How large is the normal force on the bead at point Ⓐ if its mass is 5.00 g?

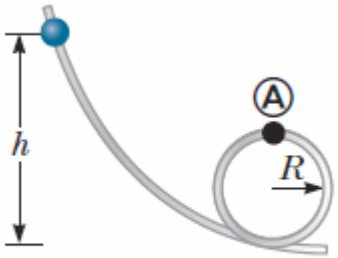
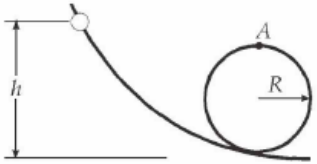
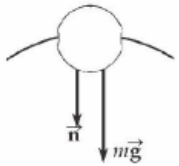


Figure P8.5

The speed at the top can be found from the conservation of energy for the bead-track-Earth system, and the normal force can be found from Newton's second law.





ANS. FIG. P8.5

(a) We define the bottom of the loop as the zero level for the gravitational potential energy.

Since $v_i = 0$,

$$E_i = K_i + U_i = 0 + mgh = mg(3.50R)$$

The total energy of the bead at point Ⓐ can be written as

$$E_A = K_A + U_A = \frac{1}{2}mv_A^2 + mg(2R)$$

Since mechanical energy is conserved, $E_i = E_A$, we get

$$mg(3.50R) = \frac{1}{2}mv_A^2 + mg(2R)$$

simplifying,

$$v_A^2 = 3.00\,gR$$

$$\boxed{v_A = \sqrt{3.00gR}}$$

(b) To find the normal force at the top, we construct a force diagram as shown, where we assume that n is downward, like mg. Newton's second law gives $\sum F = ma_c$, where a_c is the centripetal acceleration.

$$\sum F_y = ma_y: \quad n + mg = \frac{mv^2}{r}$$

$$n = m\left[\frac{v^2}{R} - g\right] = m\left[\frac{3.00gR}{R} - g\right] = 2.00mg$$

$$n = 2.00\left(5.00\times10^{-3}\text{ kg}\right)\left(9.80\text{ m/s}^2\right)$$

$$= \boxed{0.098\,0\text{ N downward}}$$

7. M Two objects are connected by a light string passing over a light, frictionless pulley as shown in Figure P8.7. The object of mass $m_1 = 5.00$ kg is released from rest at a height $h = 4.00$ m above the table. Using the isolated system model, (a) determine the speed of the object of mass $m_2 = 3.00$ kg just as the 5.00-kg object hits the table and (b) find the maximum height above the table to which the 3.00-kg object rises.

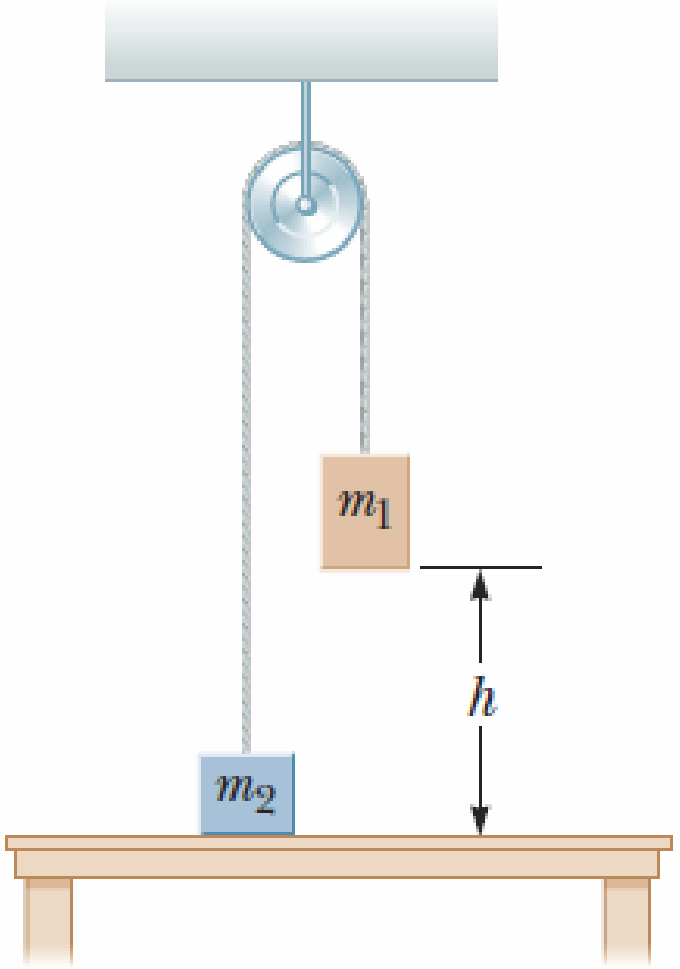
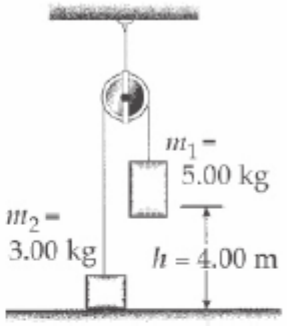


Figure P8.7
Problems 7 and 8.

We assign height $y = 0$ to the table top. Using conservation of energy for the system of the Earth and the two objects:



ANS. FIG. P8.7

(a) Choose the initial point before release and the final point, which we code with the subscript *fa*, just before the larger object hits the floor. No external forces do work on the system and no friction acts within the system. Then total mechanical energy of the system remains constant and the energy version of the isolated system model gives

$$(K_A + K_B + U_g)_i = (K_A + K_B + U_g)_{fa}$$

At the initial point, K_{Ai} and K_{Bi} are zero and we define the gravitational potential energy of the system as zero. Thus the total initial energy is zero, and we have

$$0 = \frac{1}{2}(m_1 + m_2)v_{fa}^2 + m_2gh + m_1g(-h)$$

Here we have used the fact that because the cord does not stretch, the two blocks have the same speed. The heavier mass moves down, losing gravitational potential energy, as the lighter mass moves up, gaining gravitational potential energy. Simplifying,

$$(m_1 - m_2)gh = \frac{1}{2}(m_1 + m_2)v_{fa}^2$$

$$v_{fa} = \sqrt{\frac{2(m_1 - m_2)gh}{(m_1 + m_2)}} = \sqrt{\frac{2(5.00\text{ kg} - 3.00\text{ kg})g(4.00\text{ m})}{(5.00\text{ kg} + 3.00\text{ kg})}}$$

$$= \sqrt{19.6}\text{ m/s} = \boxed{4.43\text{ m/s}}$$

(b) Now we apply conservation of energy for the system of the 3.00-kg object and the Earth during the time interval between the instant when the string goes slack and the instant at which the 3.00-kg object reaches its highest position in its free fall.

$$\Delta K + \Delta U = 0 \quad \rightarrow \quad \Delta K = -\Delta U$$

$$0 - \frac{1}{2}m_2v^2 = -m_2g\Delta y \rightarrow \Delta y = \frac{v^2}{2g}$$

$$\Delta y = 1.00\text{ m}$$

$$y_{max} = 4.00\text{ m} + \Delta y = \boxed{5.00\text{ m}}$$

12. A sled of mass m is given a kick on a frozen pond. The kick imparts to the sled an initial speed of 2.00 m/s. The coefficient of kinetic friction between sled and ice is 0.100. Use energy considerations to find the distance the sled moves before it stops.

We could solve this problem using Newton's second law, but we will use the nonisolated system energy model, here written as $-f_k d = K_f - K_i$, where the kinetic energy change of the sled after the kick results only from the friction between the sled and ice. The weight and normal force both act at 90° to the motion, and therefore do no work on the sled. The friction force is

$$f_k = \mu_k n = \mu_k mg$$

Since the final kinetic energy is zero, we have

$$-f_k d = -K_i$$

or

$$\frac{1}{2}mv_i^2 = \mu_k mgd$$

Thus,

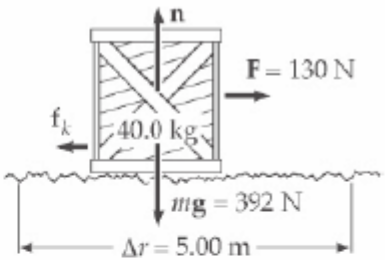
$$d = \frac{mv_i^2}{2f_k} = \frac{mv_i^2}{2\mu_k mg} = \frac{v_i^2}{2\mu_k g} = \frac{(2.00\ \text{m/s})^2}{2(0.100)(9.80\ \text{m/s}^2)} = \boxed{2.04\ \text{m}}$$

16. A 40.0-kg box initially at rest is pushed 5.00 m along a rough, horizontal floor with a constant applied horizontal force of 130 N. The coefficient of friction between box and floor is 0.300. Find (a) the work done by the applied force, (b) the increase in internal energy in the box–floor system as a result of friction, (c) the work done by the normal force, (d) the work done by the gravitational force, (e) the change in kinetic energy of the box, and (f) the final speed of the box.

$\sum F_y = ma_y$: $n - 392\ \text{N} = 0$

$n = 392\ \text{N}$

$f_k = \mu_k n = (0.300)(392\ \text{N}) = 118\ \text{N}$



ANS. FIG. P8.16

(a) The applied force and the motion are both horizontal.

$W_F = Fd\cos\theta$

$= (130\ \text{N})(5.00\ \text{m})\cos 0°$

$= \boxed{650\ \text{J}}$

(b) $\Delta E_{int} = f_k d = (118\ \text{N})(5.00\ \text{m}) = \boxed{588\ \text{J}}$

(c) Since the normal force is perpendicular to the motion,

$W_n = nd\cos\theta = (392\ \text{N})(5.00\ \text{m})\cos 90° = \boxed{0}$

(d) The gravitational force is also perpendicular to the motion, so

$$W_g = mgd\cos\theta = (392\ \text{N})(5.00\ \text{m})\cos(-90^\circ) = \boxed{0}$$

(e) We write the energy version of the nonisolated system model as

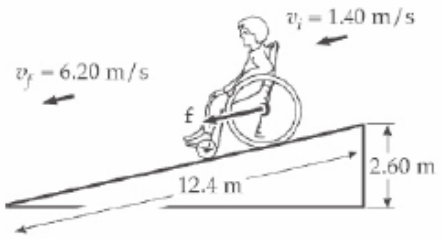
$$\Delta K = K_f - K_i = \sum W_{other} - \Delta E_{int}$$

$$\frac{1}{2}mv_f^2 - 0 = 650\ \text{J} - 588\ \text{J} + 0 + 0 = \boxed{62.0\ \text{J}}$$

(f) $$v_f = \sqrt{\frac{2K_f}{m}} = \sqrt{\frac{2(62.0\ \text{J})}{40.0\ \text{kg}}} = \boxed{1.76\ \text{m/s}}$$

19. A boy in a wheelchair (total mass 47.0 kg) has speed 1.40 m/s at the crest of a slope 2.60 m high and 12.4 m long. At the bottom of the slope his speed is 6.20 m/s. Assume air resistance and rolling resistance can be modeled as a constant friction force of 41.0 N. Find the work he did in pushing forward on his wheels during the downhill ride.

The boy converts some chemical energy in his body into mechanical energy of the boy-chair-Earth system. During this conversion, the energy can be measured as the work his hands do on the wheels.



ANS. FIG. P8.19

$$\Delta K + \Delta U + \Delta U_{body} = -f_k d$$

$$(K_f - K_i) + (U_f - U_i) + \Delta U_{body} = -f_k d$$

$$K_i + U_i + W_{hands\text{-}on\text{-}wheels} - f_k d = K_f$$

Rearranging and renaming, we have

$$\frac{1}{2}mv_i^2 + mgy_i + W_{by\,boy} - f_k d = \frac{1}{2}mv_f^2$$

$$W_{by\,boy} = \frac{1}{2}m\left(v_f^2 - v_i^2\right) - mgy_i + f_k d$$

$$W_{by\,boy} = \frac{1}{2}(47.0\text{ kg})\left[(6.20\text{ m/s})^2 - (1.40\text{ m/s})^2\right] - (47.0\text{ kg})(9.80\text{ m/s}^2)(2.60\text{ m}) + (41.0\text{ N})(12.4\text{ m})$$

$$W_{by\,boy} = \boxed{168\text{ J}}$$

29. **W** An 820-N Marine in basic training climbs a 12.0-m vertical rope at a constant speed in 8.00 s. What is his power output?

The Marine must exert an 820-N upward force, opposite the gravitational force, to lift his body at constant speed. The Marine's power output is the work he does divided by the time interval:

$$\text{Power} = \frac{W}{t}$$

$$P = \frac{mgh}{t} = \frac{(820\ \text{N})(12.0\ \text{m})}{8.00\ \text{s}} = 1\ 230\ \text{W} = \boxed{1.23\ \text{kW}}$$

30. The electric motor of a model train accelerates the train from rest to 0.620 m/s in 21.0 ms. The total mass of the train is 875 g. (a) Find the minimum power delivered to the train by electrical transmission from the metal rails during the acceleration. (b) Why is it the minimum power?

(a) $$P_{av} = \frac{W}{\Delta t} = \frac{K_f}{\Delta t} = \frac{mv^2}{2\Delta t} = \frac{(0.875\ \text{kg})(0.620\ \text{m/s})^2}{2(21\times10^{-3}\ \text{s})} = \boxed{8.01\ \text{W}}$$

(b) Some of the energy transferring into the system of the train goes into internal energy in warmer track and moving parts and some leaves the system by sound. To account for this as well as the stated increase in kinetic energy, energy must be transferred at a rate higher than 8.01 W.